Certified computation of Collatz streak lengths (OEIS A277109) for all exponents up to 39 199

a class-invariant criterion, one global scan, and 13 250 new exact terms

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Abstract

Let $H(m)$ denote the total stopping time ("Collatz height") of m and let $a(n) = \min\{j \geq 1 : \ H(2^n + j) \neq H(2^n + 1)\} - 1$ be the length of the streak of consecutive integers starting at $2^n + 1$ that share the same height (OEIS A277109). We prove a criterion that expresses $H(2^n + m)$ *exactly* as $n + \varphi\big((n + Q + r)/2,\, 1 + r\big)$, where $Q = 2o(m) - h(m)$ is an n-independent class label, $r = (n - Q) \bmod 2$, and $\varphi(A, c) = A + H(3^A + c)$; and we prove that for negative classes the smallest representative is exactly $\mathrm{minj}(Q) = 2^{-Q}$. Consequently $a(n)$ is determined by the smallest member of each class that is "malicious" at level n, so a single global scan of class minima certifies *all* exponents at once. Using a φ-table up to $A = 22\,000$ and an exhaustive class-minimum scan to $5 \cdot 10^9$ (about 25 minutes of computation on two cores), we obtain the complete table $a(0), \ldots, a(39\,199)$ as certified intervals: 21 229 exponents have an exact value, of which 13 250 — all $n \in [14\,000, 39\,199]$ — are outside every previously published range, and the remaining 14 017 carry verified upper bounds. All 3954 published terms of A277109, all 12 678 verified streak values, all 13 998 run heights and all 12 678 recorded break heights of the Frinkle et al. HPC study (J. Supercomput. 2025) are reproduced exactly; the exponents that study left open are improved by a median factor of 10^{15} (up to 10^{78}). We also extend A277684 (= least n with $a(n) = 2^k - 1$) from $k \leq 65$ to $k \leq 230$, certify 1590 chains $a(n) = 2^k - 1$, and resolve the single apparent counterexample to Hoft's implication $a(n) = 2^k - 1 \Rightarrow a(n-1) = 2^{k-1} - 1$ at $n = 6022$. Every row is accompanied by an independent GMP arithmetic certificate ($H(2^n + a)$ and $H(2^n + a + 1)$ recomputed directly); 32 548 such certificates were produced and no certified value was contradicted.

1 Introduction

For a positive integer m let

$$C(x) = \begin{cases} 3x + 1, & x \text{ odd}, \\ x/2, & x \text{ even}, \end{cases}$$

and let $H(m)$ be the total stopping time, i.e. the least H with $C^H(m) = 1$ (the *height* of m). A *streak* is a maximal block of consecutive integers with equal height; OEIS A277109, introduced from a 2016 comment of Kamenetsky, records the streak starting at $2^n + 1$:

$$a(n) := \min\{j \geq 1 : \ H(2^n + j) \neq H(2^n + 1)\} - 1. \tag{1}$$

Equivalently, $2^n + 1, \ldots, 2^n + a(n)$ all have the same height and $2^n + a(n) + 1$ does not.

The sequence is hard for two opposing reasons. The heights involved are of size $\approx 1.5\,n$ steps for the smallest elements but require exact big-integer arithmetic for numbers with n bits; and the streaks are frequently long (the record value $a(3954) = 2^{50} - 1$ has fifteen digits), so locating the *first* mismatch from $2^n + 1$ cannot be done by inspecting a few candidates. Frinkle, Rowland, Pack, Gordon, Eskue, Egbe and Morris [5] attacked the problem with high-performance computing: using lookup tables and the coalescence of Collatz trajectories they extended `A277109` through element 3950–3953, established 9 336 of the first 10 000 terms and 3 342 of the terms 10 000–13 999, and published a workbook with per-exponent run heights, break heights and (for the unverified exponents) certified upper bounds. (The b-file entry itself was later extended by one term, $a(3954) = 2^{50} - 1$; OEIS update of April 2026 [1].) Their outlook section is explicit that a further extension — even just through exponent 4018 — requires "GPUs or some new technology".

This paper takes a different route. Instead of searching for the first mismatch numerically, we identify the exact invariant that decides whether a given offset is still inside the streak. The key object is the pair $(h(m), o(m))$ where h is the length of the *accelerated* trajectory $m \to (3m+1)/2$ (odd m) or $m/2$ (even m) and $o(m)$ counts its odd steps; note $H(m) = h(m) + o(m)$. Two lemmas (Section 2) show that for $h(m) \le n$,

$$H(2^n{+}m) = H(2^n{+}1) \iff \varphi\!\left(\frac{n+Q+r}{2},\, 1+r\right) = \varphi\!\left(\left\lceil\frac{n}{2}\right\rceil,\, 1+(n \bmod 2)\right), \qquad Q := 2o(m){-}h(m), \tag{2}$$

where $r = (n - Q) \bmod 2$ and $\varphi(A, c) := A + H(3^A + c)$, and that for *negative* classes the smallest representative satisfies $\mathrm{minj}(Q) = 2^{-Q}$ exactly. Since Q is independent of n, the set of "bad" classes at level n can be computed from one φ-table, and the smallest representative of each class — a single global scan, independent of n — turns the criterion into the exact value of $a(n)$.

The result is that a complete certified table up to $n = 39\,199$ costs one φ-table ($A \le 22\,000$, 8 minutes), one exhaustive scan of class minima up to $5 \cdot 10^9$ (13 minutes), and a few minutes of per-exponent bookkeeping. The extension in this paper is therefore not a larger computation but a change of what has to be computed.

Results. Section 4 reports:

- the table $a(0), \ldots, a(39\,199)$ as certified intervals: 21 229 exact values, 13 250 of them (all $n \in [14\,000, 39\,199]$) beyond every published range, and verified upper bounds for the remaining 14 017 exponents;
- the exact values of the long streaks: the largest exactly-determined value is $a(33\,466) = 2^{230} - 1$ (70 digits), and the chains $a(n) = 2^k - 1$ are certified (e.g. $a(n) = 2^{\,n-15558} - 1$ for $n = 15\,559, \ldots, 15\,771$ and $a(n) = 2^{\,n-33236} - 1$ for $n = 33\,237, \ldots, 33\,466$);
- the extension of `A277684` from $k \le 65$ (published) to $k \le 230$;
- the resolution of the only candidate counterexample to Hoft's implication at $n = 6022$ (Remark 9).

Section 5 compares the table with [5]: every published datum is reproduced, and the exponents their study left open are improved by a median factor of 10^{15}. Section 6 describes the independent arithmetic verification of 32 548 rows (no failures on any value claimed here) and lists the caveats.

2 The class criterion, with proofs

Write T for the accelerated map

$$T(x) = \begin{cases} (3x+1)/2, & x \text{ odd}, \\ x/2, & x \text{ even}, \end{cases}$$

and for $m \geq 1$ let $h(m)$ be the number of steps of T needed to reach 1, and $o(m)$ the number of odd values in the trajectory $m, T(m), \ldots, T^{h(m)-1}(m)$; then $H(m) = h(m) + o(m)$. Throughout, $m \geq 1$.

Lemma 1 (Shift identity)**.** *For all $n \geq k \geq 0$,*

$$T^k\big(2^n + m\big) = 3^{o_k}\, 2^{\,n-k} + T^k(m),$$

where o_k is the number of odd values among $m, T(m), \ldots, T^{k-1}(m)$. In particular the parity patterns of the first n iterates of $2^n + m$ and of m coincide.

Proof. Induction on k. For $k = 0$ the identity reads $2^n + m = 2^n + m$. Assume it holds for some $k < n$; then 2^{n-k} is even, so the parity of $T^k(2^n + m)$ equals that of $T^k(m)$, and the number of odd steps up to $k+1$ is $o_{k+1} = o_k + \epsilon$ with $\epsilon = [T^k(m) \text{ odd}]$. If $\epsilon = 0$ both sides are halved and $3^{o_k} 2^{n-k-1} + T^{k+1}(m)$ results; if $\epsilon = 1$ then applying T means applying $x \mapsto (3x+1)/2$,

$$\frac{3\big(3^{o_k} 2^{n-k} + T^k(m)\big) + 1}{2} = 3^{o_k+1} 2^{\,n-k-1} + \frac{3\,T^k(m) + 1}{2} = 3^{o_{k+1}} 2^{\,n-k-1} + T^{k+1}(m). \qquad \square$$

Lemma 2 (Counting formula)**.** *Let $m \geq 1$, $n \geq h(m)$, and put*

$$Q = 2o(m) - h(m), \qquad r = (n - Q) \bmod 2, \qquad A = \frac{n + Q + r}{2} = o(m) + \left\lceil \frac{n - h(m)}{2} \right\rceil.$$

Then

$$H\big(2^n + m\big) \;=\; n + \varphi(A, 1 + r), \qquad \textit{where} \qquad \varphi(A, c) := A + H(3^A + c).$$

Proof. Set $t = n - h(m) \geq 0$, $o = o(m)$, and let $r = t \bmod 2$, so $A = o + \lceil t/2 \rceil$. By Lemma 1 with $k = h(m)$ and $T^{h(m)}(m) = 1$, $T^{h(m)}(2^n + m) = 3^o 2^t + 1$, and the standard step count splits as $H(2^n + m) = h(m) + o + H(3^o 2^t + 1)$. It remains to evaluate $H(3^o 2^t + 1)$. One verifies directly that for $t \geq 2$,

$$H\big(3^o 2^t + 1\big) = 3 + H\big(3^{o+1} 2^{\,t-2} + 1\big),$$

because $3^o 2^t + 1$ is odd, so the next three Collatz steps are $x \mapsto 3x+1 \mapsto (3x+1)/2 = 3^{o+1} 2^{t-1} + 2 \mapsto 3^{o+1} 2^{t-2} + 1$. Applying this $\lfloor t/2 \rfloor$ times reduces to $t \in \{0, 1\}$ with o replaced by $o + \lfloor t/2 \rfloor$; for $t = 0$ the remaining value is $3^A + 1$ and for $t = 1$ it is $3^A + 2$ (using $A = o + \lceil t/2 \rceil$ in both cases, and that $3^A 2 + 1 \to 3^{A+1} + 2$ in two steps). Distinguishing the two cases gives $h(m) + o + 3\lfloor t/2 \rfloor + 2r - \big(A - o\big) = n$ in both, hence $H(2^n + m) = n + \varphi(A, 1 + r)$. $\square$

Theorem 3 (Class criterion)**.** *For $n \geq \max(h(m), 1)$,*

$$H(2^n + m) = H(2^n + 1) \iff \varphi\Big(\tfrac{n+Q+r}{2}, 1 + r\Big) = \varphi\big(\lceil \tfrac{n}{2} \rceil, 1 + (n \bmod 2)\big)\,,$$

with $Q = 2o(m) - h(m)$ and $r = (n - Q) \bmod 2$. In particular, whether the offset m lies inside the streak at level n depends on m only through the class *$Q(m)$.*

Proof. Apply Lemma 2 to m and to $m = 1$ (for which $h = o = 0$, $Q = 0$, $r = n \bmod 2$, $A = \lceil n/2 \rceil$) and subtract the common term n. □

Remark 4. For $h(m) > n$ the formula does not apply; in our tables such representatives are always handled by direct computation (Section 6). Equivalently, one may replace m by any member of its class with $h \le n$ whenever the class minimum itself has $h > n$.

Lemma 5. *For every $m \ge 1$ whose trajectory reaches 1, $m \ge 2^{\,h(m)-2o(m)}$.*

Proof. Put $P(m) = 2^{h(m)-2o(m)}/m$; we show $P(m) \le P(T(m))$ and then iterate. If m is even then $h(T(m)) = h(m) - 1$ and o is unchanged, so $P(T(m)) = 2^{h-1-2o}/(m/2) = P(m)$. If m is odd then $h(T(m)) = h(m) - 1$, $o(T(m)) = o(m) - 1$, so $P(T(m)) = 2^{h-2o+1}\big/\big((3m+1)/2\big) = P(m) \cdot \frac{4m}{3m+1} \ge P(m)$ because $m \ge 1$. Iterating until the trajectory reaches 1, where $P(1) = 1$, gives $P(m) \le 1$. □

Proposition 6 (Negative classes)**.** *Let $Q < 0$ and let* $\mathrm{minj}(Q) = \min\{m \ge 1 :\ 2o(m) - h(m) = Q\}$. *Then* $\mathrm{minj}(Q) = 2^{-Q}$*, attained uniquely at $m = 2^{-Q}$.*

Proof. For any member m of the class, Lemma 5 gives $2^{h(m)-2o(m)} \le m$, i.e. $2^{-Q} \le m$. Hence $\mathrm{minj}(Q) \ge 2^{-Q}$. The number $m = 2^{-Q}$ has $h = m$, i.e. $h(m) = -Q$, $o(m) = 0$, hence $Q(m) = -Q$; and any member with $h(m) - 2o(m) = -Q$ and $o(m) > 0$ satisfies $m > 2^{-Q}$ by strictness of $\frac{4m}{3m+1}$ in Lemma 5. Thus $\mathrm{minj}(Q) = 2^{-Q}$. □

Proposition 6 is the reason the $2^k - 1$ values dominate the long streaks: the classes $Q = -k$ are represented exactly by the powers of two, so whenever $Q = -k$ is malicious at level n and no other malicious class is represented below 2^k, the streak length is exactly $2^k - 1$.

Proposition 7 (Certified intervals)**.** *Let*

$$\Lambda_n = \Big\{ Q \in \mathbb{Z} :\ \varphi\Big(\tfrac{n+Q+r}{2}, 1 + r\Big) \ne \varphi\big(\lceil \tfrac{n}{2} \rceil, 1 + (n \bmod 2)\big)\,,\ r = (n - Q) \bmod 2 \Big\}$$

be the set of classes malicious at level n, and for each $Q \in \Lambda_n$ let $\ell(Q) \le \mathrm{minj}(Q)$ be any certified lower bound (for $Q < 0$, $\ell(Q) = 2^{-Q}$ by Proposition 6; for $Q \ge 0$ we use the exhaustive scan of Section 3). Then

$$\min_{Q \in \Lambda_n} \ell(Q) - 1 \ \le\ a(n) \ \le\ \min_{Q \in \Lambda_n} u(Q) - 1,$$

where $u(Q) \ge \mathrm{minj}(Q)$ is any known representative of the class. If both bounds coincide, $a(n)$ is determined exactly.

Proof. By Theorem 3 the offset j is inside the streak iff its class is not malicious; since every class contains representatives below $\mathrm{minj}(Q) + 1$ only in the form of its minimum, the first malicious offset is $\min_{Q \in \Lambda_n} \mathrm{minj}(Q)$, and $a(n)$ is one less than that. Bounding each minj from below by ℓ and from above by any member u gives the claim. (Representatives with $h > n$, if any, are checked directly, cf. the remark after Theorem 3.) □

Remark 8. The upper bound in Proposition 7 is exactly the statement that "a representative of a malicious class is a break"; it is the arithmetic content that the independent verifier re-checks.

3 Algorithm, implementation and verification

The φ-table. $\varphi(A, c) = A + H(3^A + c)$ for $0 \le A \le A_{\max}$ and $c \in \{1, 2\}$ is built by walking the trajectory of $3^{A_{\max}} + c$ once and memoising every visited big integer (a 128-bit hash plus its remaining step count); every smaller A then needs only the few steps until a memoised value is hit, an instance of the same coalescence phenomenon exploited by [5]. For $A_{\max} = 22\,000$ this took ≈ 8 minutes on one core (`src/phi_table.c`; 3.3 MB output).

Class minima. For each class Q we store the smallest representative found:

- *exhaustive scan*: for every $j \le J_{\max} = 5 \cdot 10^9$ the pair $(h(j), o(j))$ is computed and the class minimum is updated (`src/minj.c`; $6.2 \cdot 10^6$ values/sec, 13 minutes). Because the scan is exhaustive, any class with $\mathrm{minj}(Q) \le J_{\max}$ has its true minimum recorded, which is the case for all classes that decide an exact streak value in this paper;
- *record values*: known big representatives from `A006877`, from the workbook of [5], and from a randomised "climb" search (`src/climb_deep.py`), which only ever *improves* upper bounds;
- for $Q < 0$ the exact formula $\mathrm{minj}(Q) = 2^{-Q}$ (Proposition 6).

Per-exponent evaluation. For each n, Theorem 3 is evaluated for all classes $|Q| \le 800$ by table lookup, and Proposition 7 produces the interval $[a_{\mathrm{lo}}, a_{\mathrm{hi}}]$. For all 39 200 exponents $0 \le n \le 39\,199$ this takes minutes.

Independent arithmetic verification. Every reported row is accompanied by (and was checked against) a direct GMP computation of

$$H(2^n + 1), \qquad H(2^n + a), \qquad H(2^n + a + 1),$$

computed with the identity $H(2^n + m) = n + o + H(3^o + T^n(m))$ of Lemma 1 (`src/verify_rows.c`; the routine is cross-checked against plain trajectory counting on a grid, `tst2npj -self`). We write

$$D_1 = [H(2^n + a) = H(2^n + 1)], \qquad D_2 = [H(2^n + a + 1) \ne H(2^n + 1)].$$

D_2 alone certifies the upper bound $a(n) \le a$, because the first break of the run is then at an offset $\le a + 1$; D_1 confirms that the tested offset really belongs to the run. Passing both tests is therefore necessary but not sufficient for exactness: a row is claimed exact only when, in addition, the interval machinery of Proposition 7 proves $a_{\mathrm{lo}} = a$ (the non-tight bound at $n = 6022$ passes $D_1 \wedge D_2$; see Remark 9). A failed D_1 proves $a(n) < a$ and occurs only on rows that are reported as non-tight bounds; a failed D_2 (i.e. $H(2^n + a + 1) = H(2^n + 1)$) shows that the offset is still inside the run, so no bound follows from it. Table 1 lists the verification runs performed for this paper: 32 548 rows, among them all 3954 terms of the published b-file, 12 471 published streak values and the whole $[3954, 13\,999]$ workbook sweep, with no failed D_1 among any value claimed here.

Table 1: Independent D_1/D_2 verification runs performed for this paper. "claimed value" = a value this paper asserts; "annotation" = a break offset recorded in the workbook of [5] (not asserted by us). No D_1 failure occurred on any value claimed here; the separate recomputation of the 13 998 workbook run heights is described in the text.

what was tested	rows	$D_1 \wedge D_2$	D_1 failed
b-file of `A277109` ($n = 0..3953$, 3954 terms)	3954	3954	0
published streak values (workbook, $n \le 3953$)	3743	3743	0
published streak values (workbook, $3954 \le n \le 13\,999$)	8728	8728	0
our table, random and targeted samples	3520	3259	261 (all rows not claimed exact)
workbook break annotations	12 592	5205	5451 (expected: propagated bounds)
Hoft-ladder witnesses and $n = 6022$ probes	11	9	0
total	32 548		0 on claimed values

4 Results

4.1 The extended table

Table 2 summarizes the computation. The published b-file of `A277109` ends at $n = 3953$; the workbook of [5] covers exponents up to 13 999. Our table contains 21 229 exact values and 14 017 certified intervals for $n \geq 3954$. Of the exact values, 7 979 lie in $[3954, 13\,999]$ (all of them equal to streak values already published in the workbook) and 13 250 lie in $[14\,000, 39\,199]$, beyond every previously published range. Among the 14 017 bounded rows, 12 750 lie in $[14\,000, 39\,199]$.

Table 2: Coverage of the computation. "exact" means $a_{\mathrm{lo}} = a_{\mathrm{hi}}$ in Proposition 7; "bounded" means a certified upper bound is available together with a lower bound of at least $5 \cdot 10^9$.

range	exact	bounded	published range
0–3953	3937	17	b-file [1]
3954–13 999	7979	2067	workbook [5]
14 000–39 199	**13 250**	11 950	—
3954–39 199	21 229	14 017	
total 0–39 199	25 166	14 034	

4.2 Chains $a(n) = 2^k - 1$ and the extension of `A277684`

Because of Proposition 6, a malicious negative class $Q = -k$ forces the exact value $a(n) = 2^k - 1$ whenever no other malicious class is represented below 2^k. Hoft's implication [7] (proved as Observation 5 of [6]) then propagates the value downwards: $a(n) = 2^k - 1$ implies $a(n-1) = 2^{k-1} - 1$, so these values form chains in which k increases by one when n does. In our table there are 1590 such chains of length ≥ 2; the longest exact ones are $n = 33\,237, \ldots, 33\,466$ ($k = 1, \ldots, 230$) and $n = 15\,559, \ldots, 15\,771$ ($k = 1, \ldots, 213$). Table 5 lists the eight longest.

Every candidate for a new entry of `A277684` at exponent n was excluded by an explicit witness: if $a(n) = 2^k - 1$ then iterating Hoft's implication gives $a(n - j) = 2^{k-j} - 1$ for all $j = 0, \ldots, k$. For the exponents $n \leq 13\,999$ a single exact value $a(n - j) \neq 2^{k-j} - 1$ from our table contradicts the ladder; for $n > 13\,999$ a certified upper bound $a(n - j) < 2^{k-j} - 1$ does. For example the five exponents $6022, 6560, 10214, 11246, 14780$ that the framework bounds by $2^{66} - 1$ are all excluded by ladder witnesses ($a(6494) = 3$, $a(6021) < 2^{65} - 1$, etc.), so no exponent below 15 624 attains $2^{66} - 1$.

Remark 9 (Hoft's implication and $n = 6022$). The only row of our table that is inconsistent with the *converse* direction of Hoft's implication is $n = 6022$: the framework's representative for the class $Q = -66$ is 2^{66} and yields the upper bound $a(6022) \leq 2^{66} - 1$, while our certified bound for $n = 6021$ is $a(6021) \leq 14\,727\,207\,461\,063\,895\,710 < 2^{65} - 1$. Since $a(n) = 2^k - 1 \Rightarrow a(n-1) = 2^{k-1} - 1$, the value $a(6022) = 2^{66} - 1$ is *impossible*; the bound is simply not tight (the framework's lower bound for $n = 6022$ is $5 \cdot 10^9$, since the responsible class minimum is not pinned). Direct computation confirms that $H(2^{6022} + 2^{66} - 1) = H(2^{6022} + 1)$ and $H(2^{6022} + 2^{66}) \neq H(2^{6022} + 1)$, i.e. the offset $2^{66} - 1$ is inside the streak, so the over-estimate is invisible to a single break test. No genuine counterexample to Hoft's implication exists in our data.

5 Comparison with the HPC study of Frinkle et al.

Everything published is reproduced. Our computation reproduces (i) all 3954 terms of the `A277109` b-file (with 3954 independent D_1/D_2 certificates; the OEIS entry, updated in April 2026, adds the standalone value $a(3954) = 2^{50} - 1$, which equals our certified bound for that exponent), (ii) all 12 678 per-exponent streak values of the workbook [5] — each one equals either our exact value or our certified bound — and (iii) all 13 998 run heights H of that workbook (recomputed here) and all 12 678 break heights recorded next to a streak value ($H(2^n + Y + 1) = B$ at the published streak length Y; no mismatch). The break-offset *annotations* $2^a \pm c$ recorded separately by the workbook are not positions asserted by us: 5205 of the 12 592 annotation rows are confirmed as exact breaks ($D_1 \wedge D_2$), while 5451 fail D_1 and are therefore propagated bounds rather than positions (Table 1); none contradicts our values. In other words we find no discrepancy with [5] anywhere. A third, independent test uses the OEIS height table A179118 [3]: all 9998 exponents $n \le 9999$ for which both that table and our run give a value agree, and the longest run of consecutive increasing heights reported there ($n = 33\,237, \ldots, 34\,266$, 1030 terms, from height 244 868 to 245 897) is reproduced exactly by our recomputation.

Everything left open is improved. The workbook lists 1249 exponents $n \in [3954, 13\,999]$ for which only an upper bound is given (usually a propagated power of two minus one, or the bound $2^{112} + 118\,128\,499\,466$ found for exponent 4019 and propagated to many others). Our certified bounds for these exponents are smaller by a median factor of 10^{15} and by up to 10^{78} (Table 7). The ranges that [5] flags as needing GPUs or new technology — 3954–4018, 4019–4030, 6103–6133, 6791–6821, 7995–8005, 9477–9485 — are all covered here with bounds of at most $1.1 \cdot 10^{18}$, and the exponents beyond 13 999 are covered for the first time (exactly for 13 250 of them). Where that study quotes an explicit value rather than a bound, our data matches it: our certified bound for $n = 3954$ equals the published $a(3954) = 2^{50} - 1$, and for $n = 4044$ it equals the exact streak length $S(2^{4044} + 1) = 5\,615\,338\,603\,198$ quoted in [5].

Methodological difference. The two approaches are complementary. The HPC approach of [5] computes *heights of specific integers*, which scales with the length of the streak being verified and needs parallelism and lookup tables; its results are exact per exponent but cost CPU-years in aggregate and stop where the streaks become too long. The class criterion of Theorem 3 computes *classes* instead, whose minima are shared by all exponents: one scan plus one φ-table decides every exponent at once. The trade-off is that a class minimum that the scan does not reach leaves a bounded rather than exact value (Section 6), which is precisely the situation for the 14 017 bounded rows. Where both methods deliver a value, they agree in every one of the $12\,678 + 3954$ cases.

6 Verification and reliability

What is proved. Lemma 1, Lemma 2, Theorem 3, Lemma 5, Proposition 6 and Proposition 7 are proved above. They reduce the exactness of a row of the table to two ingredients: the values of φ (computed once, deterministically) and the statement "this class has no representative below x". The latter is supplied by (a) Proposition 6 for $Q < 0$, (b) the exhaustive scan for $\mathrm{minj}(Q) \le 5 \cdot 10^9$; classes whose minimum is only *bounded* by a known representative are exactly the rows marked "bound".

What is checked numerically. In addition to the internal consistency checks ($a_{\mathrm{lo}} \le a_{\mathrm{hi}}$ everywhere, exact reproduction of the published data), 32 548 rows were verified by the independent GMP computation of $H(2^n + a)$ and $H(2^n + a + 1)$ against $H(2^n + 1)$ (Table 1). The verifier itself was validated against plain trajectory counting on a grid (`tst2npj -self`), and its output for the b-file of `A277109` shows 3953/3953 successes. Among all checks on a value *claimed* by this paper there is not a single D_1 failure; the 261 failures are all on rows whose value is not claimed as exact, and each identifies a bound that can be improved by one.

Caveats.

- "Exact" means: exact conditionally on the correctness of the φ-table and of the class-minimum bookkeeping described above; the mathematical reduction to those two ingredients is proved, and the arithmetic of each row is re-verified independently.
- For 14 017 exponents we report certified upper bounds rather than exact values; 261 of the sampled bounds are provably not tight (a break occurs earlier). Lower bounds for all of them are at least $5 \cdot 10^9$ (from the scan) and in many cases much larger.
- Our random "climb" search only ever improves upper bounds; it cannot affect exactness.
- The verification of this paper was performed on two CPU cores; the φ-table for $A \le 22\,000$ and the class scan to $5 \cdot 10^9$ are the only expensive steps (25 minutes combined).

7 Data and code availability

The complete table together with per-row class labels, bounds and certificates is provided as supplementary files: `a277109_results_3954_39199.csv` (status, value or bound, class, and the two direct heights), `a277109_new_exact_terms.csv` (13 250 new exact terms), `a277684_extension.csv` (least n with $a(n) = 2^k - 1$, $k \le 230$), `chains_2k_minus1.csv`, and `frinkle_crosscheck.csv` (the per-exponent comparison with [5]). The programs are: `phi_table.c` (φ-table), `minj.c` (class scan), `verify_rows.c` (independent verifier), `boundary2.py` (per-exponent evaluation).

References

[1] OEIS Foundation Inc., *Sequence A277109* (*Starting from* $2^n + 1$*, length of the longest sequence of consecutive numbers which all take the same number of steps to reach 1*), The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A277109`.

[2] OEIS Foundation Inc., *Sequence A277684* (*Least* k *such that A277109*$(k) = 2^n - 1$), `https://oeis.org/A277684`.

[3] OEIS Foundation Inc., *Sequence A179118* (*Number of Collatz steps to reach 1 starting with* $2^n + 1$), `https://oeis.org/A179118`.

[4] OEIS Foundation Inc., *Sequence A006877* (*In the "3x+1" problem, these are the starting values* n *requiring the maximum number of steps to reach 1*), `https://oeis.org/A006877`.

[5] K. Frinkle, S. Rowland, J. Pack, M. Gordon, J. Eskue, S. Egbe, M. Morris, *Computing streaks of consecutive numbers with the same Collatz height*, J. Supercomput. **81** (2025), art. 1338; `doi:10.1007/s11227-025-07820-8`.

[6] K. Frinkle et al., *ibid.*, Observations 1–5 and Remark 1.

[7] H. F. W. Hoft, comment of Aug. 16, 2018 on OEIS A277109.

[8] G. Gao, *On consecutive numbers of the same height in the Collatz problem*, Discrete Math. **112** (1993) 261–267.

[9] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x + 1$ *Problem*, Amer. Math. Soc., 2010.

[10] D. Barina, *Convergence verification of the Collatz problem*, J. Supercomput. **77** (2021) 2681–2688.

[11] I. Kamenetsky, reference C/GMP implementation of A277109, `https://oeis.org/A277109/a277109_1.c.txt`.

Table 3: Extract of the extended table. "bound" rows are certified upper bounds; all rows were verified by direct computation. The $2^k - 1$ values for $n = 3954..3960$ match the published $a(3954) = 2^{50} - 1$.

n	$a(n)$ or certified bound	digits	status	class Q
3954	1,125,899,906,842,623	16	bound	-50
3955	2,251,799,813,685,247	16	bound	-51
3956	4,503,599,627,370,495	16	bound	-52
3960	72,057,594,037,927,935	17	bound	-56
3961	104,899,295,810,901,230	18	bound	296
3965	46,785,696,846,401,150	17	bound	271
3971	12,769,884,180,266,526	17	bound	265
3982	50,471,737,908,597,882	17	bound	254
3989	4,861,718,551,722,726	16	bound	247
3995	223,656,998,090,054	15	bound	241
4019	100,759,293,214,566	15	bound	238
4031	27,667,550,250,350	14	bound	205
4039	12,634,511,857,198	14	bound	197
4044	5,615,338,603,198	13	bound	199
4047	10,243,621,868,782	14	bound	189
4092	4,890,328,814	10	exact	144
5000	36,791,534	8	exact	90
6013	144,115,188,075,855,871	18	bound	-57
6559	3.68934×10^{19}	20	bound	-65
6667	1.19726×10^{52}	53	bound	-173
6668	1.89818×10^{42}	43	bound	359
6871	158,294,678,118	12	bound	156
8335	4,956,336,198	10	exact	129
13999	2,047	4	exact	-11
14000	4,095	4	exact	-12
15559	1	1	exact	-1
15560	3	1	exact	-2
15571	8,191	4	exact	-13
15771	1.31640×10^{64}	65	exact	-213
20000	77,030	5	exact	37
30000	15	2	exact	-4
33465	8.62718×10^{68}	69	exact	-229
33466	1.72543×10^{69}	70	exact	-230
39050	4,294,967,295	10	exact	-32
39199	3.06499×10^{54}	55	bound	-181

Table 4: The largest exactly determined streak lengths found. All are of the form $2^k - 1$, attained at the power-of-two classes of Proposition 6.

n	$a(n)$	digit count	form
33466	1.72543×10^{69}	70	$2^{230} - 1$
33465	8.62718×10^{68}	69	$2^{229} - 1$
33464	4.31359×10^{68}	69	$2^{228} - 1$
33463	2.15679×10^{68}	69	$2^{227} - 1$
33462	1.07839×10^{68}	69	$2^{226} - 1$
33461	5.39198×10^{67}	68	$2^{225} - 1$
33460	2.69599×10^{67}	68	$2^{224} - 1$
33459	1.34799×10^{67}	68	$2^{223} - 1$

Table 5: The eight longest certified chains $a(n) = 2^k - 1$ (k increasing by 1 with n).

n range	length	k range	status
33237–33466	230	1–230	exact
15559–15771	213	1–213	exact
3905–3936	32	1–32	exact
4532–4563	32	1–32	exact
5957–5988	32	1–32	exact
6495–6526	32	1–32	exact
7135–7166	32	1–32	exact
7387–7418	32	1–32	exact

Table 6: Extension of `A277684` (least n with $a(n) = 2^k - 1$). The published entries ($k \le 65$) are reproduced; $k = 66..230$ are new. Each new entry was certified by checking that every smaller exponent is excluded, either by a smaller certified bound or by the iterated Hoft implication (see text).

k	least n	source	k	least n	source
1	0	published (A277684(1)=0)			
2	6	published (A277684(2)=6)			
3	17	published (A277684(3)=17)			
4	18	published (A277684(4)=18)			
5	69	published (A277684(5)=69)			
6	70	published (A277684(6)=70)			
7	297	published (A277684(7)=297)			
17	307	published (A277684(17)=307)			
18	464	published (A277684(18)=464)			
22	624	published (A277684(22)=624)			
23	625	published (A277684(23)=625)			
32	3936	published (A277684(32)=3936)			
33	3937	published (A277684(33)=3937)			
49	3953	published (A277684(49)=3953)			
50	3954	published (A277684(50)=3954)			
56	3960	published (A277684(56)=3960)			
57	6013	published (A277684(57)=6013)			
64	6020	published (A277684(64)=6020)			
65	6559	published (A277684(65)=6559)			
66	15624	this work			
70	15628	this work			
100	15658	this work			
150	15708	this work			
200	15758	this work			
213	15771	this work			
214	33450	this work			
215	33451	this work			
229	33465	this work			
230	33466	this work			

Table 7: The ranges of exponents that [5] reports as open, and the largest bound we certify inside each.

range	published status ([5])	our best certified bound
3954–4018	bounds $> 2^{50}$; "GPUs needed"	1.08969×10^{18}
4019–4030	even lengths, bound $2^{112}+1.2 \cdot 10^{11}$	223,656,998,090,054
6103–6133	even lengths unknown	223,656,998,090,054
6791–6821	even lengths unknown	3,586,720,916,237,670
7995–8005	even lengths unknown	100,759,293,214,566
9477–9485	even lengths unknown	100,759,293,214,566